SDG 4 · Quality Education

DIGITAL SAFETY TOOLKIT FOR SCHOOLS ("CYBER-HYGIENE" KIT)

TEAM: CYBERSECURITY & TECHNICAL PROJECT MANAGEMENT

GROUP: CAPSONE GROUP 42

Tech4Dev

The Digital Guardian.

PROBLEM STATEMENT

Schools providing students with internet-connected devices face a critical gap in digital safety. Three core vulnerabilities define the problem: students rely on weak, reused passwords that make their accounts easy targets; invisible trackers and malicious advertisements follow students across websites, harvesting personal data without consent; and teachers, though willing, lack the practical knowledge or tools to create a digitally secure classroom. Together, these gaps mean that every time a student goes online to learn, they are exposed to preventable harm.

SDG ALIGNMENT

The SDG 4 calls for inclusive, equitable and quality education for all. In a world where learning is increasingly digital, this goal cannot be achieved without a safe digital environment. If students cannot browse freely without being tracked, manipulated by advertisements, or having their accounts compromised, the promise of "Quality Education" is hollow. Cyber hygiene is therefore not a technical add-on but a *prerequisite* for realizing SDG 4 in digital-first classrooms.

PROPOSED SOLUTION

The proposed solution is a standardized Digital Safety Toolkit; a practical, easy-to-deploy "Digital Survival Kit" that makes online safety automatic and accessible for both students and educators. It is built on three pillars:

The Password Manager (e.g., Bitwarden): Generates and stores complex, unique credentials for every student account, eliminating the risk of reused or weak passwords and preventing account takeovers that compromise student identities and academic records.

The Browser Safety Extensions: (e.g., uBlock Origin, HTTPS Everywhere). Lightweight browser add-ons that automatically block advertisements, trackers and malicious scripts: creating a clean, distraction-free and private learning environment without requiring any action from the student.

The Privacy Checklists and Training Guides: Plain-language, step-by-step guides that empower non-technical teachers to independently secure social media settings, school applications and simple training materials that build lasting cyber-awareness in both students and staff.

Success Criteria: The toolkit will be piloted in 1–2 schools, with success defined as ≥70% of users demonstrating improved cyber awareness via pre/post survey, measurable teacher adoption in classroom routines, a reduction in reported account and data incidents post-deployment.

TEAM ROLES

Technical Project Management Team (TPM): The Technical Project Management team is responsible for the overall planning, coordination, and successful delivery of the Digital Safety Toolkit project. They develop the project schedule, define scope, and break down tasks into manageable components for execution. They monitor project progress, track deliverables and ensure timelines are met while addressing any risks or challenges that may arise during execution. Additionally, the TPM team oversees documentation, resource allocation and quality assurance to ensure that the final solution is practical, user-friendly and meets the required standards.

Cybersecurity Team: The cybersecurity team ensured that all tools in the Digital Safety Toolkit meets strong security standards and protect users from online threats.

They also developed guidelines for identifying phishing, safe usage, including password management and data privacy practices for students and staff.

AI COMPONENT

- ChatGPT will be used as a drafting and ideation assistant, helping the team structure content, refine documentation, and articulate project concepts clearly.
- Claude will support in-depth research and analysis, enabling us to explore relevant literature, trends, and insights to strengthen the project's foundation.
- Gemini will assist in problem discovery and definition, helping the team identify key challenges and refine the core problem statement.
- Gamma will be used to translate the technical framework into a polished, visually compelling presentation, ensuring the final proposal is professional, engaging, and easy to understand.